
SurgiCore Pro

Enterprise Surgical Management Platform

PROJECT CASE STUDY

Precision in Every Procedure

Client	SurgiCore Pro (Internal Product)
Industry	Healthcare / Surgical Operations Management
Platform	Web Application (SPA)
Duration	Full-Stack Enterprise Build
Status	Production Ready

Executive Summary

SurgiCore Pro is an enterprise-grade surgical management platform designed for multi-specialty hospitals and surgical centers. The platform provides a comprehensive suite of tools for managing operating room scheduling, surgical case tracking, patient management, billing, inventory, staff coordination, and deep analytics — all within a single, unified web application.

Built with a modern React-based architecture and a professional, clinically sophisticated design system, SurgiCore Pro addresses the critical operational challenges faced by surgical centers: OR underutilization, scheduling conflicts, compliance tracking, and fragmented data across multiple systems.

The platform serves both the public-facing marketing site and the internal operational dashboard, creating a seamless experience from lead generation to daily surgical operations management.

200+ Surgical Centers Trusted	17+ Routes & Pages	8 Dashboard Modules	9 Public Pages
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Technology Stack

Category	Technology	Purpose
Frontend Framework	React 18 + TypeScript 5	Component-based UI architecture
Build Tool	Vite 5	Lightning-fast HMR and optimized builds
Styling	Tailwind CSS v3	Utility-first design system with semantic tokens
UI Components	shadcn/ui + Radix UI	Accessible, enterprise-grade component library
Animation	Framer Motion + GSAP	Smooth, professional micro-interactions
Charts	Recharts	Data visualization for surgical analytics
Routing	React Router DOM v6	Nested layouts with protected routes
State Management	React Context + TanStack Query	Auth state and server state management
Form Handling	React Hook Form + Zod	Type-safe form validation
Icons	Lucide React	Consistent, professional icon system
HTTP Client	Axios	API communication layer

Design System

The platform implements a comprehensive design system built on HSL-based semantic color tokens, ensuring consistent theming across all components. The color palette centers around Deep Teal (#0F766E) as the primary color, paired with Sterling Cyan (#14B8A6) as accent, creating a professional, clinically sophisticated visual identity.

Token	Value	Usage
Primary	#0F766E (Deep Teal)	Buttons, links, active states
Secondary/Accent	#14B8A6 (Sterling Cyan)	Highlights, gradients
Background	#F8FAFC	Page backgrounds
Foreground	#0F172A	Text, dark elements
Muted	#F1F5F9	Subtle backgrounds, borders
Destructive	#DC2626	Error states, alerts

Success	#16A34A	Positive indicators
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Application Architecture

Routing & Layout System

The application uses a nested layout architecture with React Router v6. Two primary layouts serve distinct user experiences:

- **PublicLayout** — Wraps all marketing pages with Header, Footer, ScrollToTop, GoToTop button, and a timed ScheduleModal popup.
- **DashboardLayout** — Wraps protected dashboard routes with a collapsible sidebar, top bar with search/notifications, and authentication guard.

Route	Page	Layout
/	Home (Landing Page)	Public
/about	About SurgiCore Pro	Public
/services	Platform Services	Public
/doctors	Medical Team	Public
/blog	Insights & Resources	Public
/book	Book Consultation	Public
/contact	Contact Us	Public
/staff-login	Staff Authentication	Public
/dashboard	Overview & KPIs	Dashboard
/dashboard/surgeries	Surgery Management	Dashboard
/dashboard/schedule	OR Scheduling	Dashboard
/dashboard/patients	Patient Directory	Dashboard
/dashboard/billing	Billing & Revenue	Dashboard
/dashboard/staff	Staff Management	Dashboard
/dashboard/inventory	Inventory & Sterilization	Dashboard
/dashboard/analytics	Surgical Analytics	Dashboard
/dashboard/settings	System Settings	Dashboard

Public-Facing Website

The public website serves as the primary conversion funnel for SurgiCore Pro. Every page is designed with enterprise-level professionalism, strong CTAs, trust signals, and conversion-optimized layouts.

Home Page

The landing page features a full-viewport hero section with a cinematic surgical command center background image, dark overlay with floating KPI stat cards, animated trust signals, and dual CTAs. A compliance bar showcases HIPAA, ISO 27001, SOC 2, 99.99% Uptime SLA, and 256-bit Encryption certifications.

Key Components Across Public Pages

- **Header** — Sticky transparent-to-solid navigation with scroll detection, mobile hamburger menu with Framer Motion animation
- **Footer** — 4-column layout with platform links, company links, contact info, and compliance badges
- **ScheduleModal** — localStorage-persistent popup triggered after 5 seconds, offering a free 15-minute surgical operations strategy call
- **GoToTop** — Animated floating button that appears after 400px scroll
- **ScrollToTop** — Instant scroll-to-top on every route change
- **SharedSections** — Reusable TrustBadges, StatsSection, TestimonialsSection, and CTASection components

Conversion Features

- Timed modal popup for lead capture (5-second delay, shown once per visitor)
- Multiple CTA touchpoints: Book Consultation, Request Demo, Schedule Free Call
- Trust signals on every page: HIPAA, ISO 27001, SOC 2 badges
- Social proof: 200+ surgical centers, 2,400+ surgeons, 99.2% satisfaction rate
- Professional testimonials from recognizable healthcare institutions

Dashboard Platform

The dashboard provides a comprehensive surgical operations management interface with 8 dedicated modules, each containing rich data visualizations, interactive tables, quick-action modals, and real-time status indicators.

Module	Key Features
Overview	Surgical KPI cards, OR utilization trend charts (Recharts), today's surgery schedule, surgeon leaderboard, equipment status
Surgeries	Complete surgery registry with advanced filtering, status management (Scheduled/Pre-Op/In Progress/Completed)
Schedule	Multi-OR calendar view with list toggle, drag-and-drop scheduling concept, surgeon and anesthesia availability heatmap
Patients	Surgical patient directory with pre/post-op filters, patient surgical history, consent management, pre-op assessment
Billing	Procedure-based billing, insurance claims tracker with status pipeline, profit/loss by procedure type, revenue breakdown
Staff	Surgeon, anesthetist, and OT staff management, cross-theater scheduling, performance metrics, credential and certification
Inventory	Surgical instruments, implants, and consumables tracking, sterilization cycle management, expiry alerts, stock level
Analytics	Case duration trend analysis (12-month), outcome distribution charts, surgeon benchmarking with comparative metrics

Dashboard UI Components

- **Collapsible Sidebar** — Framer Motion animated sidebar with icon-only collapsed state, smooth width transitions
- **Top Bar** — Logo, global search input, notification bell, user profile with role display, live/mock data toggle
- **Authentication Guard** — React Context-based auth with redirect to /staff-login for unauthenticated users
- **Mock Data Engine** — Comprehensive realistic surgical data: 8+ surgery records, 6 patients, 5 surgeons, 6 inventory items, billing records, and 30-day trend data

SEO & Branding

Search Engine Optimization

- Complete meta tags: title, description, Open Graph (og:title, og:description, og:image, og:type)
- Twitter Card meta tags for social sharing with branded OG image (1200x630)
- Semantic HTML structure with single H1 per page
- Custom branded favicon (teal medical cross PNG)
- Apple touch icon for iOS home screen
- Professional page titles under 60 characters

Brand Identity

The entire platform is branded exclusively as SurgiCore Pro with zero third-party or generator branding. The brand identity includes:

- Custom SVG logo mark: teal rounded rectangle with medical cross icon (Activity icon from Lucide)
- Typography: Plus Jakarta Sans for headings, Inter for body text
- Branded OG image with logo, tagline, and HIPAA/SOC2/ISO27001 trust badges
- Custom favicon matching brand colors
- Consistent brand voice: enterprise, trustworthy, clinical sophistication

Responsive Design

The entire application is fully responsive with mobile-first design principles. The layout uses a 90vw centered container with a max-width of 1480px. Breakpoints follow Tailwind CSS defaults (sm: 640px, md: 768px, lg: 1024px, xl: 1280px). The dashboard sidebar collapses to icon-only mode on smaller screens, and the public header transforms into a hamburger menu.

Project File Structure

Directory/File	Description
src/pages/	9 public pages + 9 dashboard pages
src/components/	Header, Footer, GoToTop, ScrollToTop, ScheduleModal, SharedSections, DashboardSidebar, Dashb
src/components/ui/	50+ shadcn/ui components (Button, Card, Dialog, Table, Tabs, etc.)
src/layouts/	PublicLayout.tsx, DashboardLayout.tsx
src/contexts/	AuthContext.tsx for authentication state
src/data/	mockData.ts with comprehensive surgical mock data
src/hooks/	use-mobile.tsx, use-toast.ts
src/assets/	Hero background, blog images (6 professional medical images)
src/lib/	utils.ts with cn() helper
public/	favicon.png, og-image.jpg, robots.txt
index.html	Complete meta tags, OG tags, favicon
tailwind.config.ts	Custom theme with surgical color palette
src/index.css	Design tokens, custom utilities, scrollbar hiding

Conclusion & Key Outcomes

SurgiCore Pro represents a complete, production-ready enterprise surgical management platform that demonstrates mastery across full-stack web development, enterprise UI/UX design, and healthcare domain expertise.

Key Technical Achievements

- **17+ fully functional routes** with nested layouts, protected routes, and seamless navigation
- **50+ reusable UI components** built on shadcn/ui with custom enterprise styling
- **Comprehensive design system** with HSL-based semantic tokens, dark mode support, and consistent spacing
- **Rich data visualization** using Recharts with real-time surgical KPIs and trend analysis
- **Professional animations** using Framer Motion for smooth page transitions and micro-interactions
- **Enterprise-grade security UX** with authentication context, protected routes, and HIPAA compliance messaging
- **Conversion-optimized marketing** with timed modals, trust signals, and multi-touchpoint CTAs
- **Full responsive design** from mobile to ultra-wide with adaptive layouts
- **Complete SEO implementation** with meta tags, OG images, structured data readiness, and semantic HTML
- **Custom branding** with generated images, favicon, OG social share image, and zero third-party branding

"SurgiCore Pro exemplifies what enterprise healthcare software should look and feel like — trustworthy, sophisticated, and built for the demands of modern surgical operations."